

Experiment 4: Monitor AWS Resources Using CloudWatch

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Objective

- Set up CloudWatch to monitor metrics and logs
- Create alarms for proactive monitoring

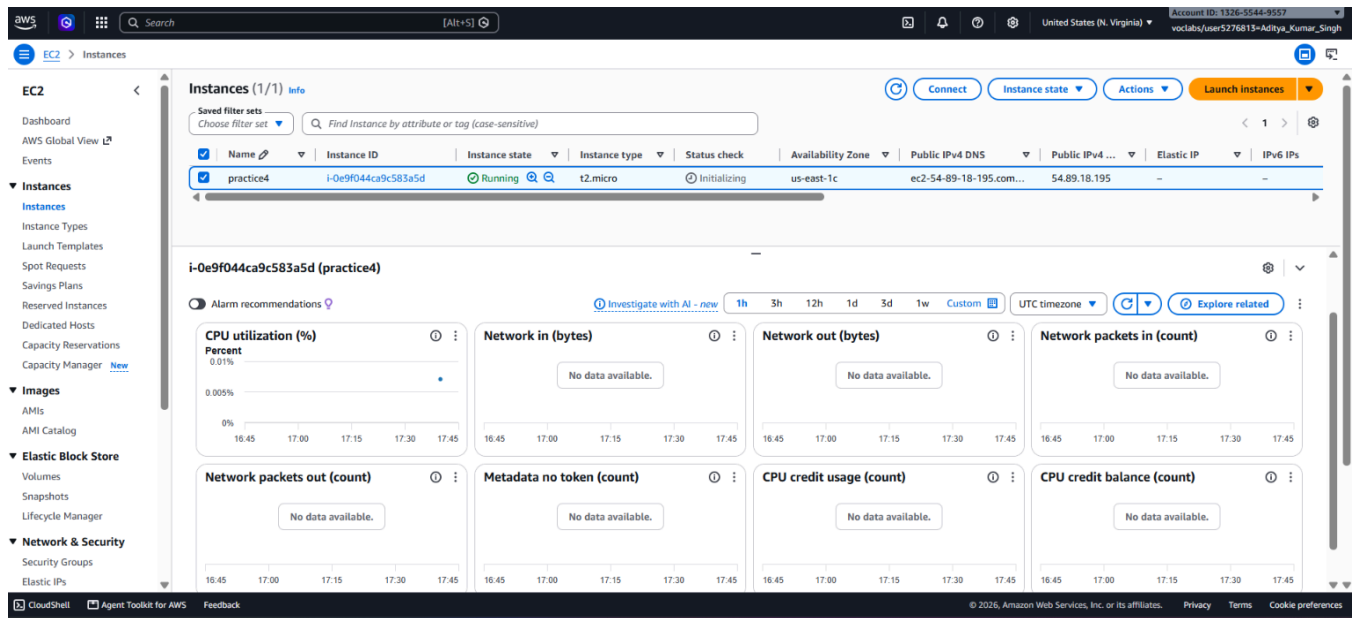
Introduction

Amazon CloudWatch is a monitoring and observability service that collects metrics, logs, and events from AWS resources and applications, enabling teams to track performance and respond to operational issues in real time. It supports customizable alarms, log analysis, and dashboards for a unified view of system health. This experiment demonstrates enabling EC2 monitoring, creating alarms for CPU utilization, installing the CloudWatch Agent for custom metrics and log collection, and visualizing resource data through a dashboard.

Procedure

1. Enable Monitoring for EC2

- Navigated to the EC2 Monitoring tab and verified basic metrics (CPU utilization, network activity) with 5-minute granularity.



2. Create a CloudWatch Alarm

- Created a CloudWatch alarm for EC2 CPU Utilization with a threshold greater than 70%.
- Configured an SNS topic for email notifications, with an evaluation period of 2 out of 3 datapoints.

The screenshot shows the AWS CloudWatch Alarms console in the us-east-1 region. A green banner at the top indicates "Successfully created alarm High-CPU-Usage-Alarm." Below this, a blue banner states "Some subscriptions are pending confirmation" with a link to "View SNS Subscriptions". The left sidebar shows the CloudWatch navigation menu. The main content area displays a table of alarms with one entry: "High-CPU-Usage-Alarm" in a "Warning" state. The conditions for this alarm are "CPUUtilization > 70 for 2 datapoints within 15 minutes".

ClouwatchALarmCreationWorking.mp4

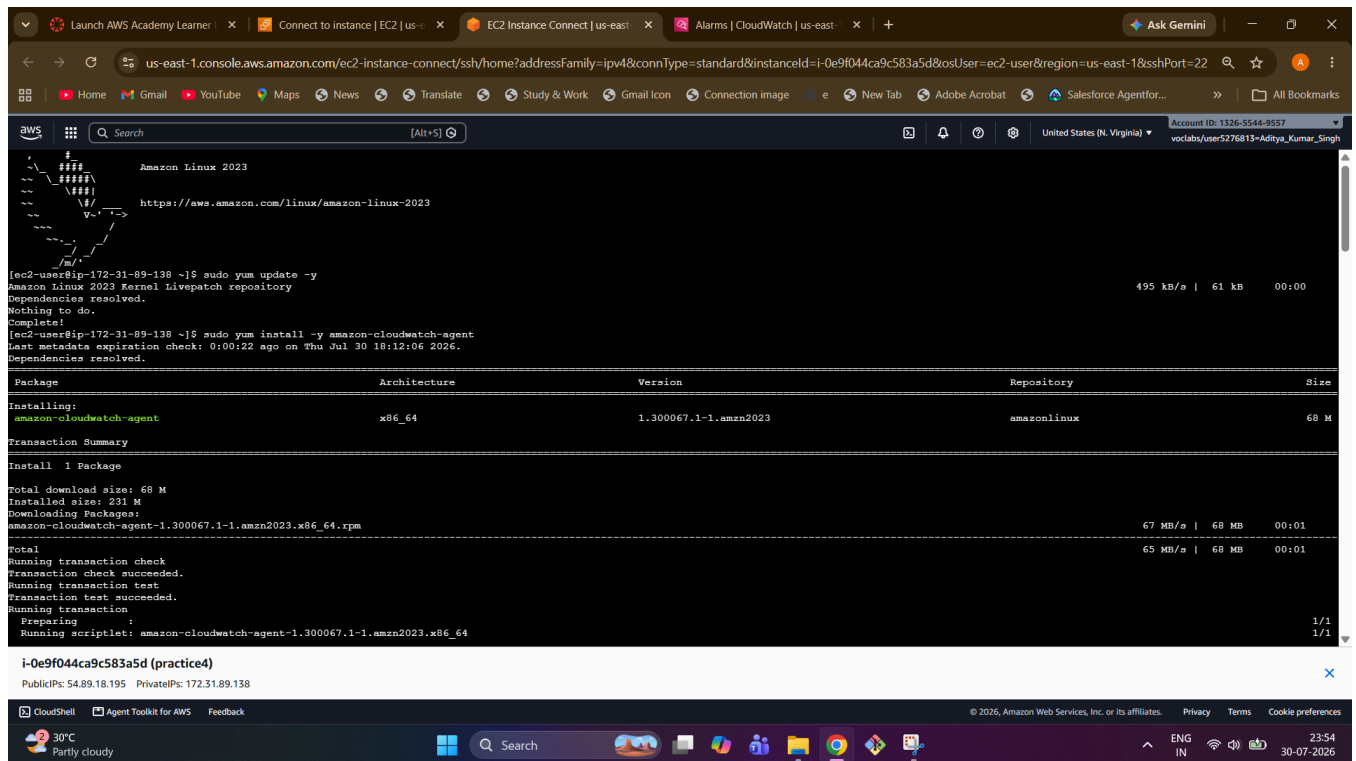
The screenshot shows the AWS CloudWatch Overview dashboard. The top navigation bar includes filters for "1h", "3h", "12h", "1d", "1w", and "Custom". The "Alarms by AWS service" section shows a bar chart for "EC2" with a legend indicating "In alarm 0", "Insufficient data 0", and "OK 1". The "Recent alarms" section displays a line graph for the "High-CPU-Usage-Alarm" showing CPU utilization over time. The "Application Monitoring" section includes a "Real User Monitoring" (RUM) section with a "Top 5" button and a "View all" link. A "Create application monitor" button is also visible.

3. Install CloudWatch Agent and Analyze Logs

- Installed the amazon-cloudwatch-agent package on the EC2 instance.
- Ran the configuration wizard and started the agent service to collect custom metrics (memory, disk usage) and stream system logs.

- Viewed system logs (/var/log/messages and /var/log/secure) in the CloudWatch Logs section under the configured log groups.

InstallAndConfigureCloudWatchAgent.mp4



```

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-89-138 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-89-138 ~]$ sudo yum install -y amazon-cloudwatch-agent
Last metadata expiration check: 0:00:22 ago on Thu Jul 30 18:12:06 2026.
Dependencies resolved.

Package Architecture Version Repository Size
Installing:
amazon-cloudwatch-agent x86_64 1.300067.1-1.amzn2023 amazonlinux 68 M

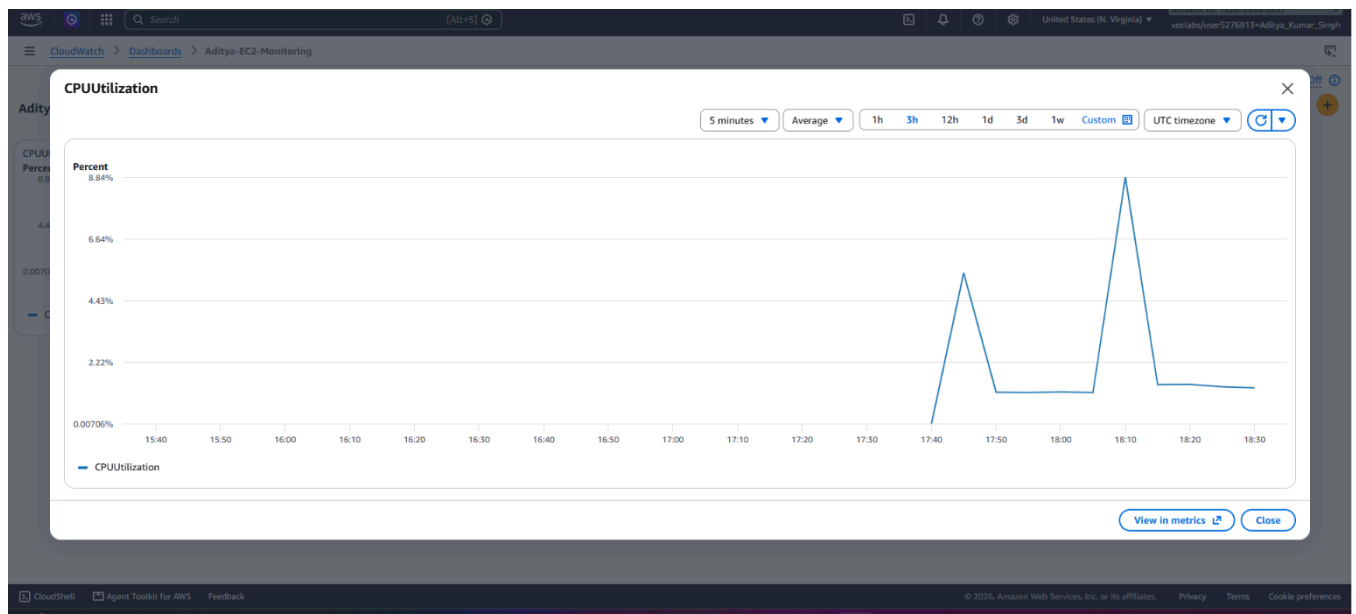
Transaction Summary
Install 1 Package

Total download size: 68 M
Installed size: 231 M
Downloading Packages:
amazon-cloudwatch-agent-1.300067.1-1.amzn2023.x86_64.rpm 67 MB/s | 68 MB 00:01
Total 65 MB/s | 68 MB 00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Running scriptlet: amazon-cloudwatch-agent-1.300067.1-1.amzn2023.x86_64
  
```

i-0e9f044ca9c583a5d (practice4)
PublicIP: 54.89.18.195 PrivateIPs: 172.31.89.138

4. Create CloudWatch Dashboard

- Created a custom dashboard with line chart widgets displaying CPU utilization, memory usage percentage, and disk usage metrics.



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United States (N. Virginia)

Account ID: 1326-5344-9557
voclabs/user5276813-Aditya_Kumar_Singh

CloudWatch > Dashboards

Custom Dashboards

Automatic dashboards

Custom Dashboards (1) info

Share dashboard Delete Create dashboard

Filter dashboards

Name	Sharing	Favorite	Last update (UTC)
Aditya-EC2-Monitoring		☆	2026-07-30 18:37

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us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#dashboards/dashboard/Aditya-EC2-Monitoring

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Account ID: 1326-5344-9557
voclabs/user5276813-Aditya_Kumar_Singh

CloudWatch > Dashboards > Aditya-EC2-Monitoring

Aditya-EC2-Monitoring ☆

Autosave: Off

1h 3h 12h 1d 3d 1w Custom UTC timezone Actions Save

CPUUtilization

Percent

8.84%

4.43%

0.00706%

16:00 17:00 18:00

CPUUtilization

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